

Rossmoyne Senior High School

Semester One Examination, 2018

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section Two:

Calculator-assumed

SOLUTIONS

Time allowed for this section

Reading time before commencing work:

ten minutes

Working time:

one hundred minutes

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators approved for use in this examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	52	35
Section Two: Calculator-assumed	12	12	100	98	65
Total					100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet.
3. You must be careful to confine your response to the specific question asked and to follow any instructions that are specified to a particular question.
4. Supplementary pages for the use of planning/continuing your answer to a question have been provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
5. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
6. It is recommended that you do not use pencil, except in diagrams.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section Two: Calculator-assumed

65% (98 Marks)

This section has **twelve (12)** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 8

(7 marks)

A research group was investigating whether the rate at which solid objects of similar volume tend to lose heat is related to their surface area.

The group had a cylinder of radius 6.3 cm and height 27 cm, a sphere of radius 9.3 cm and a cube with all sides of length 15 cm.

- (a) Show that the volume of the sphere is close to that of the cube.

(3 marks)

Solution
$V_{cube} = 15^3 = 3\,375 \text{ cm}^3$ $V_{sp} = \frac{4}{3} \times \pi \times 9.3^3$ $V = 3\,369 \text{ cm}^3$
Specific behaviours
✓ volume of cube ✓ substitutes into sphere volume formula ✓ evaluates volume of sphere

Deduct one mark for entire paper if correct units are not shown (consider Q8 & Q13)

- (b) Determine by how much the surface area of the cube exceeds the surface area of the cylinder.

(4 marks)

Solution
$S_{cube} = 15 \times 15 \times 6 = 1350 \text{ cm}^2$ $S_{cyl} = 2\pi \times 6.3(6.3 + 27)$ $S_{cyl} = 1318 \text{ cm}^2$ $\text{Diff} = 1350 - 1318 = 32 \text{ cm}^2$
Specific behaviours
✓ TSA of cube ✓ substitutes into cylinder TSA formula ✓ evaluates cylinder TSA ✓ calculates difference

Question 9

(6 marks)

- (a) Evaluate $\frac{143}{2\sqrt{11}}$, rounding your answer to the nearest whole number. (2 marks)

Solution
$\frac{143}{2\sqrt{11}} = 21.558$ Ans: 22
Specific behaviours
✓ evaluates ✓ rounds as required

- (b) Calculate the value of the expression $x^2 + y^2 + 3x - 5y$ when $x = -3.5$ and $y = 4.5$. (2 marks)

Solution
$(-3.5)^2 + (4.5)^2 + 3(-3.5) - 5(4.5) = -0.5$
Specific behaviours
✓ substitutes ✓ evaluates correctly

- (c) Use the formula $s = ut + \frac{1}{4}at^2$ to calculate the value of s when the values of the other pronumerals are $a = -8$, $t = 7.5$ and $u = 27.5$. (2 marks)

Solution
$s = 27.5 \times 7.5 + \frac{1}{4} \times (-8) \times 7.5^2$ $s = 93.75$
Specific behaviours
✓ substitutes ✓ evaluates

Question 10

(9 marks)

A person who qualifies for a government allowance will be paid \$552.30 per fortnight, so long as they do not work. In any fortnight that they do work and earn over \$230, their allowance will be reduced by 50 cents in the dollar for earnings over \$230.

- (a) A student who qualifies for the allowance earns \$280 in a fortnight. Calculate the amount of allowance they will receive. **(2 marks)**

Solution
$(280 - 230) \times 0.50 = 50 \times 0.50 = \25 reduction $552.30 - 25 = \$527.30$
Specific behaviours
✓ calculates reduction ✓ calculates net allowance

- (b) Another student who qualifies for the allowance has a part-time job for 16 hours each week, earning \$15.45 per hour. Calculate

- (i) their fortnightly wage. **(2 marks)**

Solution
$16 \times 15.45 \times 2 = \494.40
Specific behaviours
✓ calculates wage

- (ii) the amount that the fortnightly allowance is reduced. **(2 marks)**

Solution
$494.40 - 230 = 264.40$ $264.40 \times 0.5 = \$132.20$
Specific behaviours
✓ indicates excess ✓ calculates reduction

- (iii) the fortnightly sum of their allowance and wage. **(3 marks)**

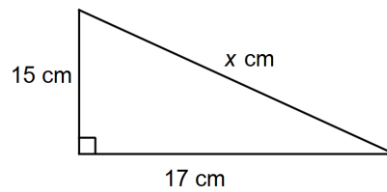
Solution
$552.30 - 132.20 = 420.10$ $420.10 + 494.40 = \$914.50$
Specific behaviours
✓ ✓ calculates allowance ✓ calculates sum

Question 11

(8 marks)

(a) Determine the value of x , correct to two decimal places, in each of the following triangles.

(i)



(2 marks)

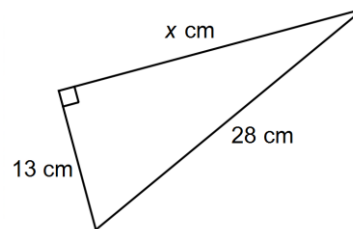
$$x^2 = 17^2 + 15^2 = 514$$

$$\therefore x = 22.67$$

✓

✓

(ii)



(3 marks)

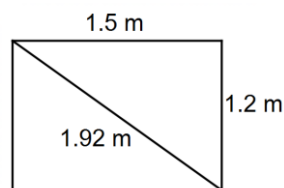
$$x^2 + 13^2 = 28^2 \rightarrow x^2 = 615$$

$$\therefore x = 24.80$$

✓✓

✓

(b) Old MacDonald had a farm, and on that farm he had a gate, as shown



He measures the length, the width and the diagonal.

He wants to know whether the gate is "square". That is, whether it is right angled.

Use Pythagoras' Theorem to determine whether the gate is "square".

(3 marks)

Assume his measurements are as correct as his measuring tape will allow.

$$1.2^2 + 1.5^2 = 3.69$$

$$1.92^2 = 3.69$$

✓

Since the sum of the squares of the shorter sides = longest side²

✓

Then right angled

✓

Question 12

(9 marks)

A grocery store has the same brand of instant coffee for sale in three different sized jars. The 75, 205 and 340 gram jars are priced at \$6.75, \$17.75 and \$29.05 respectively.

- (a) Calculate the price per 100 grams for each jar of coffee.

(3 marks)

Solution
$6.75 \div 75 \times 100 = \9.00 $17.75 \div 205 \times 100 = \8.66 $29.05 \div 340 \times 100 = \8.54
Specific behaviours
✓ divides cost by weight for at least one jar ✓ at least one price per 100 g correct ✓ all unit prices

- (b) Assuming price is the only consideration, list the jars in order of value, best value first.

(1 mark)

Solution
340g, 205g, 75g
Specific behaviours
✓ correct listing

- (c) The store is considering stocking a 480 gram jar, priced at \$43.45.

- (i) Determine how many of the existing jars are better value than this jar.

(2 marks)

Solution
$43.45 \div 480 \times 100 = \9.05 All 3 of the existing jars are better value
Specific behaviours
✓ unit price ✓ states number

- (ii) Would your answer to (i) change if the price of the 480 g jar was decreased by 5%? Justify your response.

(3 marks)

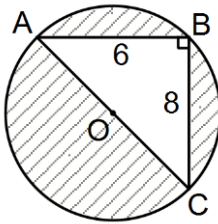
Solution
New price: $43.45 \times 0.95 = 41.28$ New unit price: $41.28 \div 480 \times 100 = \8.60 Yes - answer changed, as just 1 jar now better value than this one.
Specific behaviours
✓ changed price ✓ unit cost ✓ conclusion with reason

Question 13

(9 marks)

- (a) AC is the diameter of the circle centre O. Lengths are in cms.
Calculate the area of the shaded region.

(5 marks)



$$AC^2 = 6^2 + 8^2 = 100 \rightarrow AC = 10 \quad \checkmark$$

$$\therefore \text{Area of Circle} = \pi \times 5^2 = 78.54 \text{ cm}^2 \quad \checkmark$$

$$\therefore \text{Area of } \triangle ABC = \frac{1}{2} \times 6 \times 8 = 24 \text{ cm}^2 \quad \checkmark$$

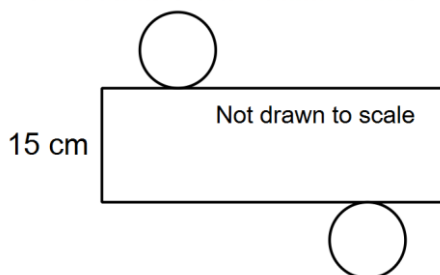
$$\text{Hence, Shaded region} = 78.54 - 24 = 54.54 \text{ cm}^2 \quad \checkmark \checkmark$$

- (b) Below is the net of a solid cylinder, which has a radius of 6 cm.

(i) Calculate the volume of the cylinder.

(2 marks)

$$V = \pi \times 6^2 \times 15 = 1696.46 \text{ cm}^3 \quad \checkmark \checkmark$$



- (ii) Calculate the surface area of the cylinder.

(2 marks)

$$SA = 2 \times \pi \times 6^2 + 2 \times \pi \times 6 \times 15 \quad \checkmark$$

$$= 791.68 \text{ cm}^2 \quad \checkmark$$

Question 14

(9 marks)

- (a) \$32 500 is invested in a 90-day term deposit offering simple interest of 3.55% pa.
Assume 365 days per year.

Calculate the value of the investment at the end of the 90 days.

(3 marks)

Solution
$I = 32500 \times \frac{3.55}{100} \times \frac{90}{365} = 284.49$ $FV = 32500 + 284.49 = \$32\,784.49$
Specific behaviours
✓ substitutes correctly ✓ evaluates interest ✓ future value, to nearest cent

- (b) A sum of \$12 500 is invested at 4.2% per annum for three years.

- (i) Determine the value of the investment after three years if the interest is compounded annually.

(2 marks)

Solution
$FV = 12500 \times \left(1 + \frac{4.2}{100}\right)^3 = \$14\,142.08$
Specific behaviours
✓ substitutes correctly ✓ evaluates, to nearest cent

- (ii) Calculate the additional interest earned over three years if the interest is compounded twice a year rather than annually.

(4 marks)

Solution
$FV = 12500 \times \left(1 + \frac{4.2}{2 \times 100}\right)^{3 \times 2} = 14160.04$ $\text{Extra} = 14160.04 - 14142.08 = \17.96
Specific behaviours
✓✓ adjusts for compounding twice per year then f/t ✓ evaluates future value ✓ extra interest, to nearest cent

Question 15**(8 marks)**

A scale model of a rectangular-based pyramid has a height of 12 cm.

- (a) Calculate the scale factor of the model, given that the real pyramid has a height of 72 cm.

(2 marks)

Solution
$72 \div 12 = 6$ Scale factor of model is $\frac{1}{6}$.
Specific behaviours
✓ obtains ratio of heights ✓ states correct scale factor

- (b) Given that

- (i) the length of the diagonal of the base of the scale model is 9.4 cm, calculate the length of the diagonal of the base of the real pyramid. **(2 marks)**

Solution
$9.4 \times 6 = 56.4 \text{ cm}$
Specific behaviours
✓✓ correct length

- (ii) the total surface area of the real pyramid is 7 236 cm², calculate the total surface area of the scale model.

(2 marks)

Solution
$6^2 = 36$ $7236 \div 36 = 201 \text{ cm}^2$
Specific behaviours
✓ indicates area scale factor ✓ correct TSA

- (iii) the volume of the scale model is 155 cm³, calculate the volume of the real pyramid.

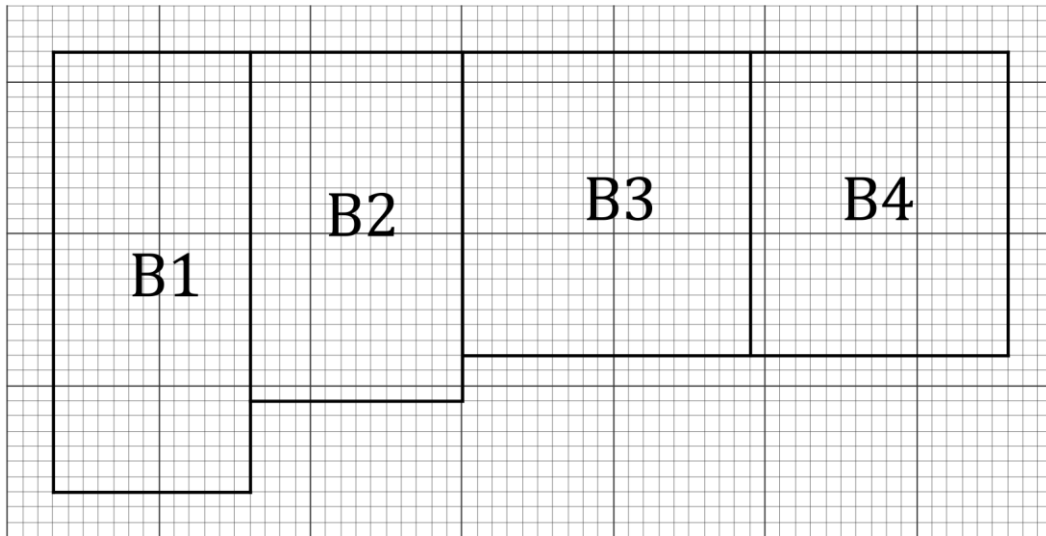
(2 marks)

Solution
$6^3 = 216$ $155 \times 216 = 33\,480 \text{ cm}^3$
Specific behaviours
✓ indicates volume scale factor ✓ correct volume

Question 16

(9 marks)

The diagram below, on 2 mm grid paper, shows a scale drawing of the floor plan of four small storage areas in the basement of an apartment block, drawn with a scale of 1 to 50.



Use dimensions from the scale drawing to answer the following questions about the actual storage areas in the apartment.

- (a) Determine the perimeter, in metres, of storage area B1. (3 marks)

Solution
Dimensions = 29 by 13 sqs = 2.9 by 1.3 m
$P = 2.9 + 2.9 + 1.3 + 1.3 = 8.4 \text{ m}$
Specific behaviours
✓ dimensions, ✓ uses scale, ✓ perimeter, correct units

- (b) Calculate the area, in square metres, of storage area B3. (2 marks)

Solution
Dimensions = 2.0 by 1.9 m
Area = $2.0 \times 1.9 = 3.8 \text{ m}^2$
Specific behaviours
✓ dimensions, ✓ area, in correct units

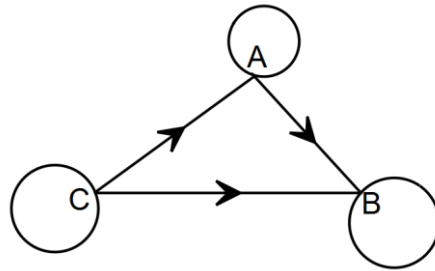
- (c) Apartment owners can rent any of the storage areas at a cost of \$26.50 per square metre per month. Determine the cost to rent storage area B2 for 15 months. (4 marks)

Solution
Dimensions = 2.3 by 1.4 m
Area = $2.3 \times 1.4 = 3.22 \text{ m}^2$
Cost = $3.22 \times 26.50 = \$85.33$
Total cost = $85.33 \times 15 = \$1\,279.95$
Specific behaviours
✓ dimensions, ✓ area, ✓ monthly cost, ✓ annual cost

Question 17

(7 marks)

The network below shows possible paths (two way and one way) between vertices A, B and C.



- (a) Complete the matrix which represents the one stage routes shown on the network. (3 marks)

		To		
		A	B	C
From	A	2	1	0
	B			
	C			

		To		
		A	B	C
From	A	2	1	0
	B	0	2	0
	C	1	1	2

✓✓✓

-1 per incorrect element in these matrices. Can give one mark in b) for squaring the matrix

- (b) Hence, or otherwise, determine the two stage route matrix for the same network. (4 marks)

		To (In two stages)		
		A	B	C
From	A			
	B			
	C			

		To (In 2 stages)		
		A	B	C
From	A	4	4	0
	B	0	4	0
	C	4	5	4

✓✓✓✓

See next page

Question 18

(9 marks)

A person invested a sum of \$42 000 for a period of six months, depositing half in a simple interest account paying 2.26% pa and using the remainder to buy shares in a small mining company for \$3.50 each.

- (a) Calculate the amount of interest earned in the deposit account.

(3 marks)

Solution
$P = 42000 \div 2 = 21000$ $I = \frac{21000 \times 2.26 \times 0.5}{100} = \237.30
Specific behaviours
✓ uses correct principle and time ✓ indicates use of simple interest formula ✓ correct interest

- (b) Ignoring any costs such as brokerage and so on, determine how many shares the person bought in the company.

(1 mark)

Solution
$21000 \div 3.50 = 6\ 000$
Specific behaviours
✓ number of shares

During the six months, the company paid the person a dividend of 5.4 cents per share.

- (c) Calculate the dividend they received.

(1 mark)

Solution
$6000 \times 0.054 = \$324$
Specific behaviours
✓ calculates dividend

- (d) At the end of the six months, the person withdrew the principal and interest from the simple interest account and sold the shares for \$3.40 each. Calculate the total gain or loss this person made on their investment of \$42 000.

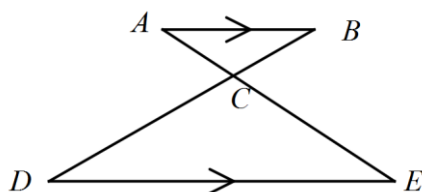
(4 marks)

Solution
$\text{Share loss} = 3.50 - 3.40 = 10 \text{ c per share}$ $\text{Loss} = 6000 \times 0.10 = 600.00$ $\text{Loss} = 600 - 237.30 - 324 = \38.70
Specific behaviours
✓ calculates loss in share value ✓ indicates use of interest and dividend, ✓ and loss ✓ correct amount, clearly stating overall loss

Question 19

(8 marks)

Two triangles are shown below. AB is parallel to DE.



- (a) Using an appropriate Condition of Similarity of triangles, outline a proof showing triangle ABC is similar to triangle EDC. (5 marks)

$$\angle A = \angle E \text{ and } \angle B = \angle D \quad \checkmark \checkmark$$

alternate angles ✓

or angle ACB = angle ECD, vertically opposite angles (instead of one of the two above)

$$\therefore \triangle ABC \sim \triangle EDC \text{ (A.A.)} \quad \checkmark \checkmark$$

First mark for correct order of vertices, second for AA condition

$$AB : ED = 1 : 2$$

- (b) If AC has length of 4 units, determine the length of AE. (3 marks)

$$\frac{AB}{ED} = \frac{AC}{EC} = \frac{1}{2} \quad \text{(Demonstrated use of ratio of 2:1)} \quad \checkmark$$

$$\frac{4}{EC} = \frac{1}{2}$$

$$\therefore EC = 8 \quad \checkmark$$

$$\therefore AE = 12 \quad \checkmark$$

Supplementary page

Question number: _____

Supplementary page

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